EXPERIMENT 6

Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

Subject: No-Code Application Development

Aim

To study cloud computing service models and explore representative IaaS, PaaS and SaaS offerings through free or trial-based services.

Objectives

- Define IaaS, PaaS and SaaS and distinguish management responsibility.
- Identify suitable application scenarios for each model.
- Examine free-tier restrictions, resource management and security considerations.

Selected Platform and Services

An available cloud provider such as AWS, Azure or Google Cloud for IaaS/PaaS exploration; a browser-based productivity tool illustrates SaaS. Free-tier eligibility, usage quotas and card requirements change by provider.

Requirements / Tools

- Internet connection and access to service documentation or account
- Optional free-tier cloud account, subject to eligibility and guardian/institution approval when necessary
- A simple sample web application or service description

No-Code Connection

Most no-code systems are delivered as SaaS, while integrations may connect to PaaS APIs and IaaS-hosted databases. The service model determines who configures servers, security, application logic and access.

Illustrative Business Scenario

Model a school event-registration app: the builder is SaaS, an optional function endpoint is PaaS, and a self-managed virtual machine is IaaS.

Experimental Procedure

Follow these steps in the designated simulator, browser or visual app builder. Menu locations and available features may vary by product version and account plan.

1. Review the provider’s free-tier eligibility, quotas, pricing and account requirements before enabling resources.

2. Locate the virtual-machine/compute offering (IaaS) and note settings such as image, CPU, storage and networking.

3. Locate an application hosting or managed runtime offering (PaaS) and note supported languages and deployment options.

4. Open a SaaS application and identify the functions available directly through the browser.

5. Compare who manages operating systems, application code, updates and physical hardware.

6. If authorized to run a free-tier resource, stop and delete it after observing its dashboard; confirm no billable resources remain.

No-Code Application Perspective

- Identify the user action and the visual interface element involved (form, page, workflow or report).
- Trace the data movement from input to processing to stored record, report or notification.
- Identify configuration that replaces traditional hand-written code and note where scripting might still be required.
- Check whether the experiment can be repeated using sample records and documented settings.

Evidence to Collect

- Screenshot of the configured screen, topology, tool or application.
- A short observation table showing input, action and output.
- One successful test and one failed or exceptional test with expected behavior.

Comparison of cloud service models

Aspect	IaaS	PaaS	SaaS
Customer controls	OS, runtime, app	Application, data	Configuration, data
Provider manages	Physical infrastructure	Infrastructure and platform	Entire hosted application
Example	Virtual machine	Managed app hosting	Online office suite
Use case	Custom servers	Rapid app delivery	Ready-to-use tools

Evaluation criteria

Criterion	Observation to record
Cost	Quota, trial expiry, possible charges
Deployment	Manual server setup versus managed platform
Scaling	Available scaling settings
Security	IAM roles, access control, data protection

Interpretation / Analysis

- IaaS offers flexibility but also responsibility for operating-system maintenance and hardening.
- PaaS reduces infrastructure administration so teams can focus on application logic.
- SaaS is simplest for end users but allows less control over underlying implementation.
- A free tier does not guarantee zero cost; usage caps and billing terms should always be checked.

Evaluation Against a No-Code Solution

- Ease of creation: determine how much is configured visually rather than programmed.
- Data and integration: identify where records, connections or services are created and managed.
- Governance: consider role permissions, traceability and safe data handling.
- Limitations: identify any feature that may require paid access, external services or custom logic.

Verification / Test Cases

Suggested Test Record

Test	Input / action	Expected observation
Normal test	Valid sample values and supported action	Expected successful output
Boundary test	Missing value, invalid input or empty data	Validation or clear empty state
Access test	Unapproved user attempts restricted action	Access limited according to role
Repeatability	Re-run same valid example	Consistent output or explain differences

Observation Notes

During the practical session, replace the illustrative outcomes with actual observed results. Include the software/platform name, version if known, screenshots, and any errors encountered.

Results

IaaS, PaaS and SaaS were compared through their control boundaries, uses and management duties. Actual console observations should be added if accounts are used.

Conclusion

The preferred cloud model depends on the required level of control, development effort and operational responsibility. PaaS often suits rapid application deployment, while SaaS offers immediate functionality.

No-Code Learning Outcome

This experiment links the technical concept to a practical no-code application development workflow. It highlights visual configuration, user experience, data handling and safe validation rather than treating the exercise as conventional full-code software development.

Precautions

- Use only fictional sample records for classroom demonstrations; avoid exposing passwords, tokens, employee information or real contact details.
- Record your own screenshots, observed outputs and deviations after executing the experiment. The examples below are models, not verified live results.
- Check access rights before sharing or publishing an app, report, workflow or externally reachable service.